Aswath Damodaran

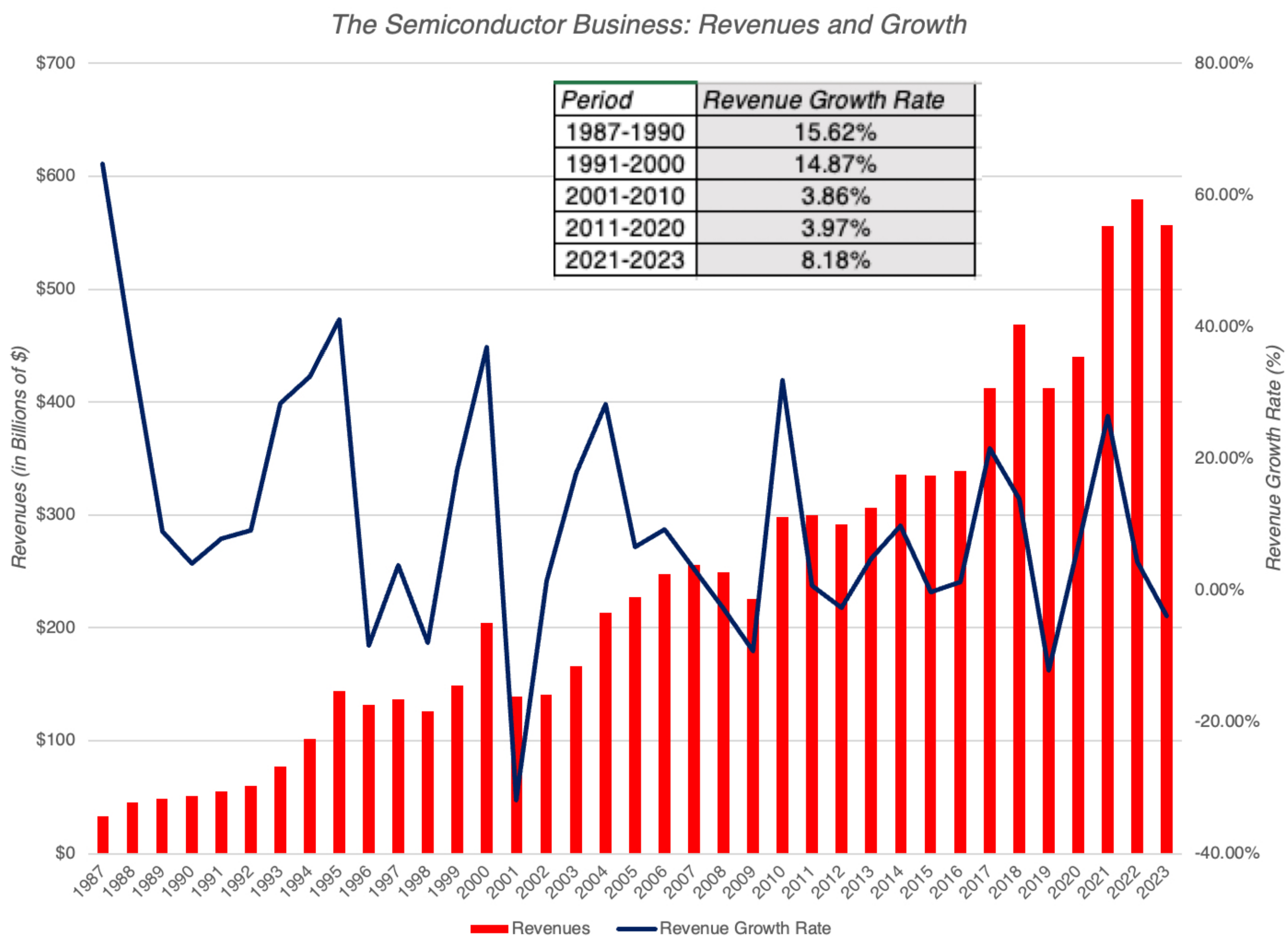
AI'S WINNERS, LOSERS AND WANNABES: VALUING NVIDIA!

ChatGPT did this!

Investment Confessions: An (Unexpected) AI Boost!

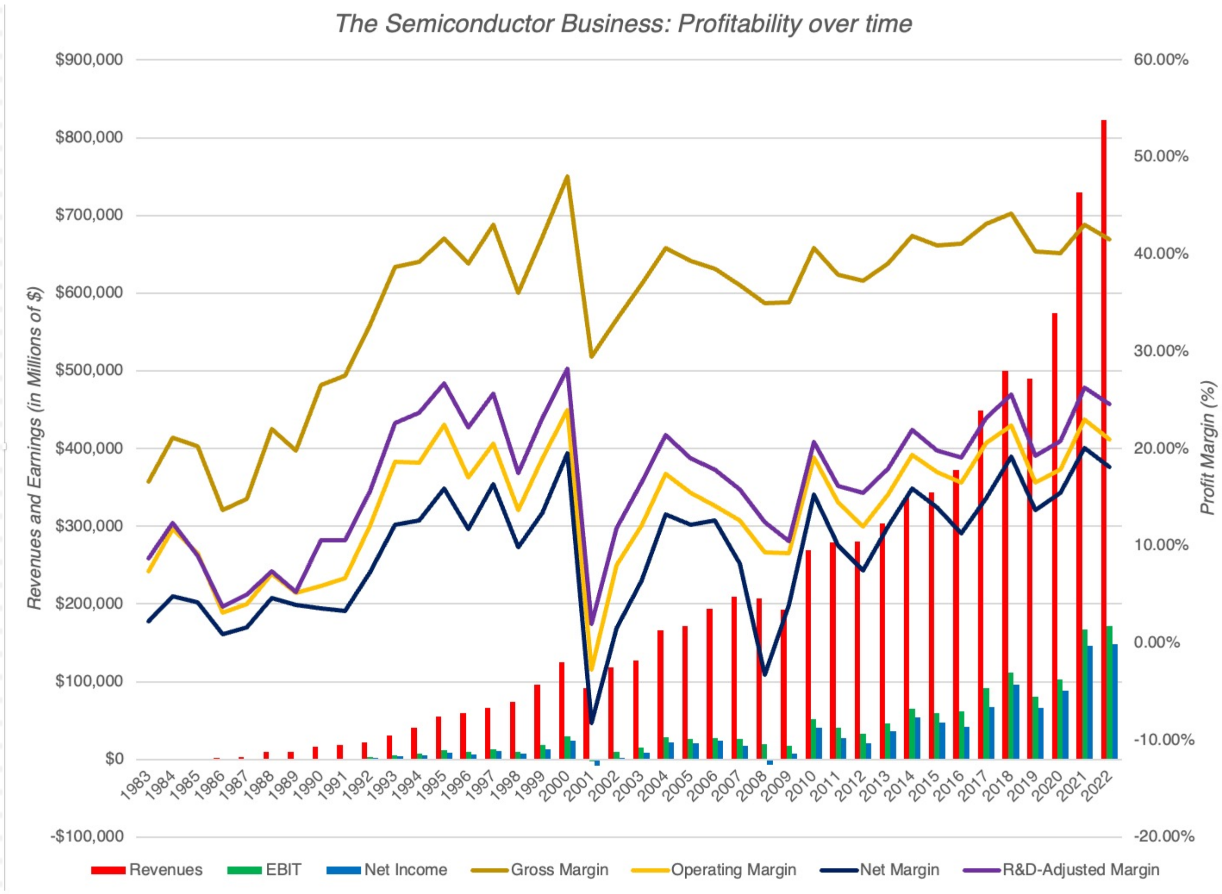
- The first is that my portfolio has held up well this year, in a market that has been top-heavy and tech-driven, and one big reason is that it contains both NVIDIA and Microsoft, two companies that have benefited from the AI story.
- The second is that much as I would like to claim credit for foresight and forward thinking, AI was not even a speck in my imagination when I bought these stocks (Microsoft in 2014 and NVIDIA in 2018).
  - I just happened to be in the right place at the right time, a reminder again that being lucky often beats being smart, at least in markets.
  - That said, NVIDIA's soaring stock price has left me facing that question of whether to cash out, or let my money ride, and thus requires an assessment of how the promise of AI play's out in its value.

1. The Semiconductor Business – A Growth Business that is maturing!



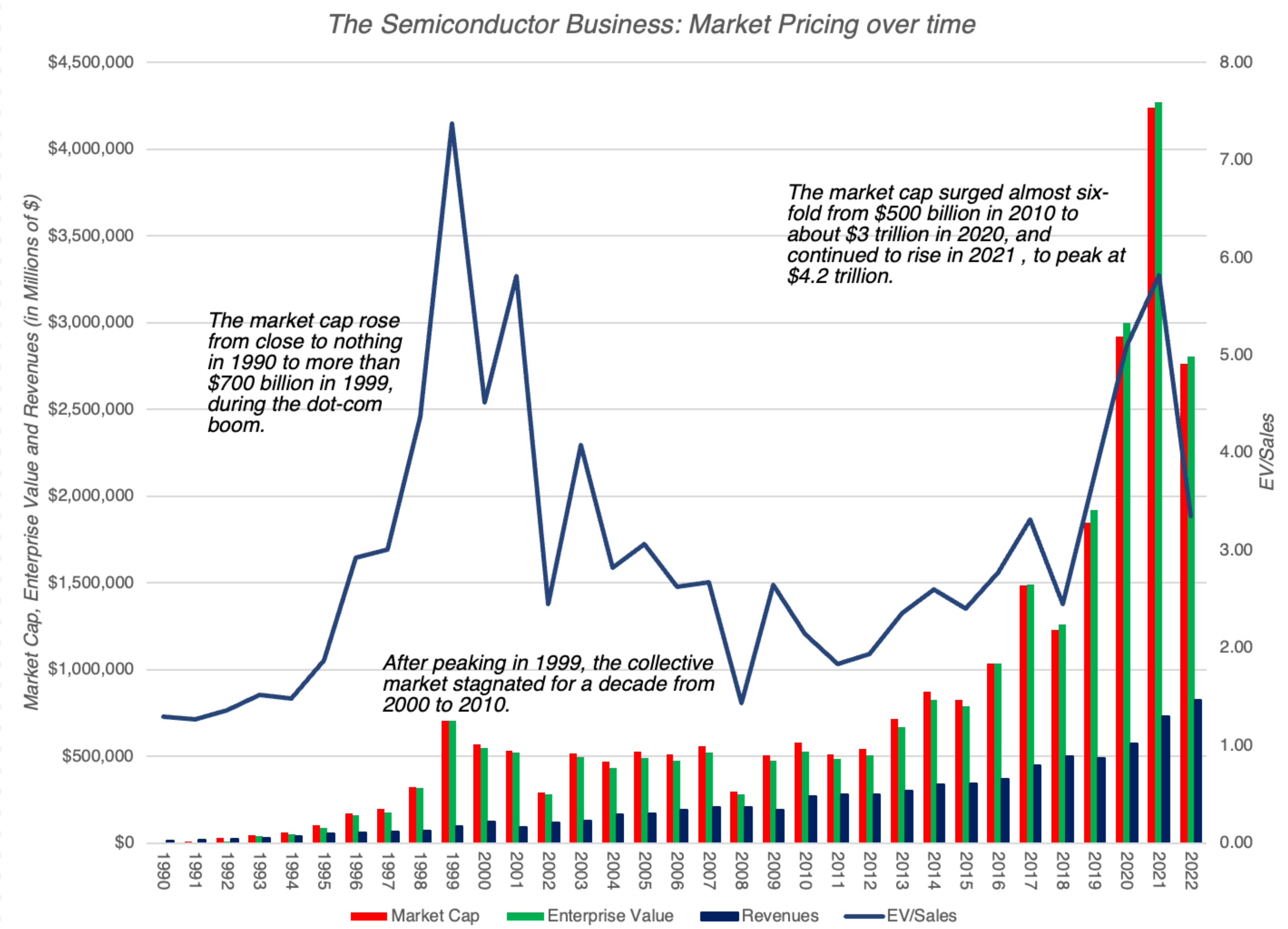
Period	Revenue Growth Rate
1987-1990	15.62%
1991-2000	14.87%
2001-2010	3.86%
2011-2020	3.97%
2021-2023	8.18%

2. Profitable, with Cycles….



Time Period	1983-1990	1991-2000	2001-2010	2011-2020	2021-2022
Gross Margin	19.36%	38.77%	36.57%	40.58%	42.28%
Operating Margin	6.70%	17.25%	11.45%	17.23%	21.94%
Net Margin	3.23%	12.22%	6.17%	13.38%	19.08%
R&D Adjusted Margin	7.73%	21.60%	14.79%	19.90%	25.46%

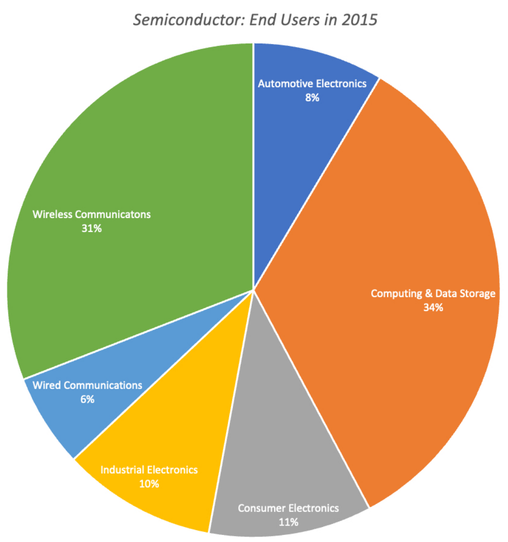
3. A Love-Hate Relationship with Markets

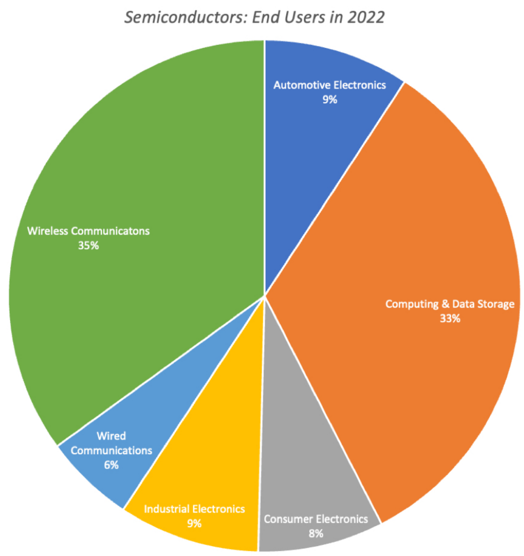


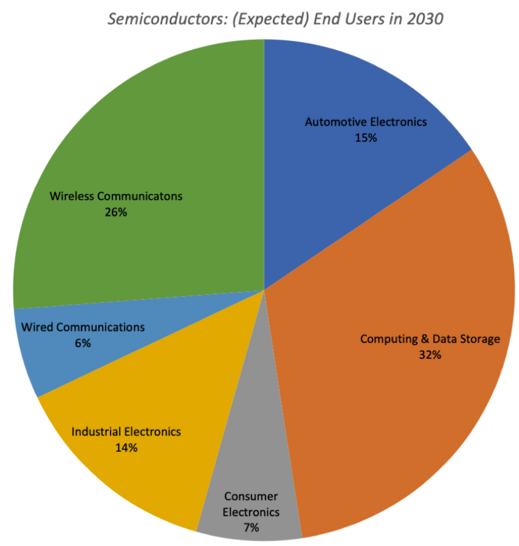
4a. With a Shifting Cast of Winners

1990	2000	2010	2020	2022
1. NEC	1. Intel	1. Intel	1. Intel	1. TSMC
2. Toshiba	2. Toshiba	2. Samsung	2. Samsung	2. Samsung
3. Hitachi	3. NEC	3. TSMC	3. TSMC	3. Intel
4. Intel	4. Samsung	4. Texas Instruments	4. SK Hynix	4. Qualcomm
5. Motorola	5. Texas Instruments	5. Toshiba	5. Micron	5. SK Hynix
6. Fujitsu	6. Motorola	6. Renesas	6. Qualcomm	6. Broadcom
7. Mitsubishi	7. STMicro	7. Qualcomm	7. Broadcom	7. Micron
8. Texas Instruments	8. Hitachi	8. STMicro	8. NVIDIA	8. NVIDIA
9. Philips	9. Infineon	9. Hynix	9. Texas Instruments	9. Applied Materials
10. Panasonic	10. Philips	10. Micron	10. Infineon	10. Texas Instruments

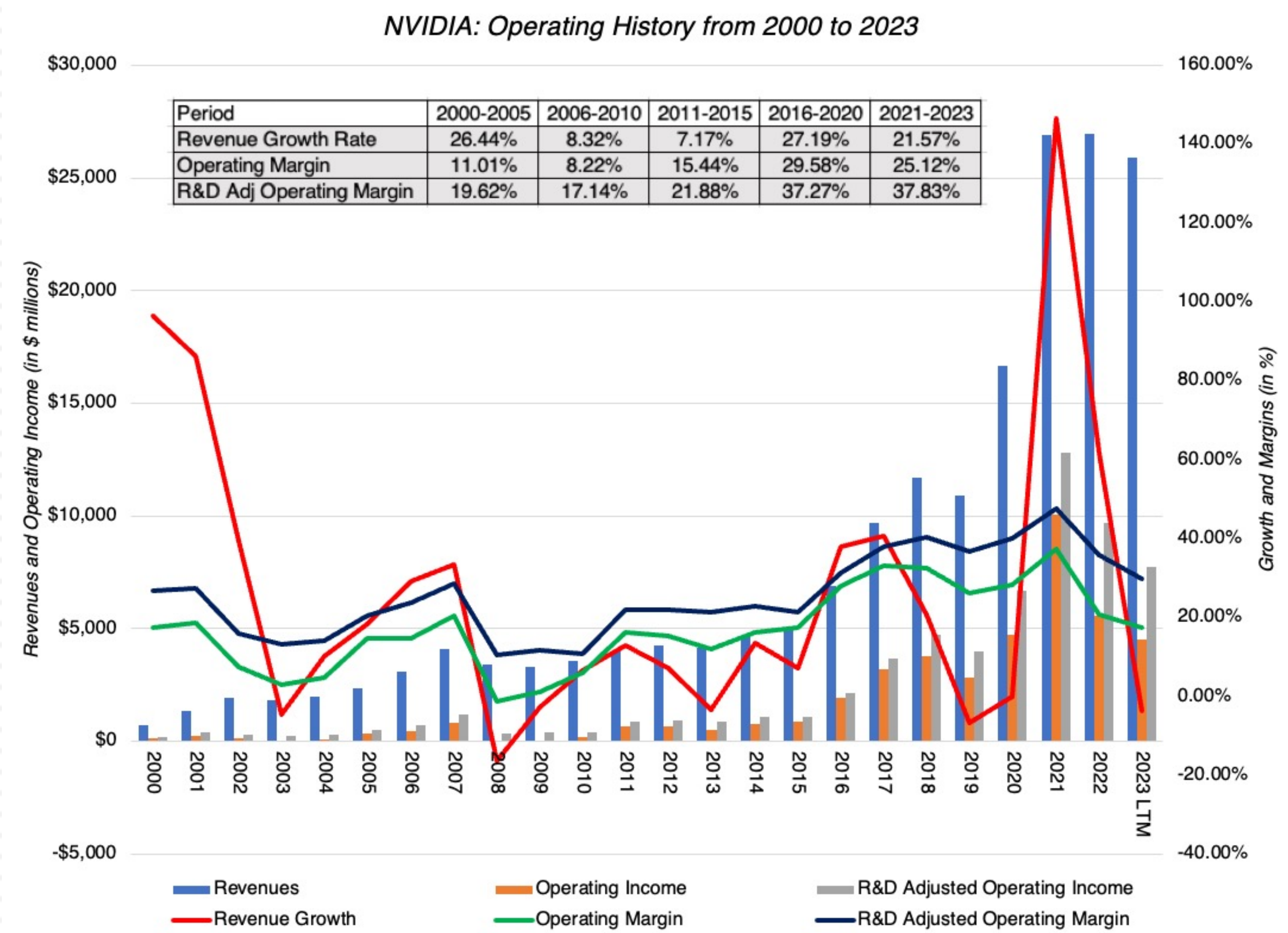
4b. And End Users



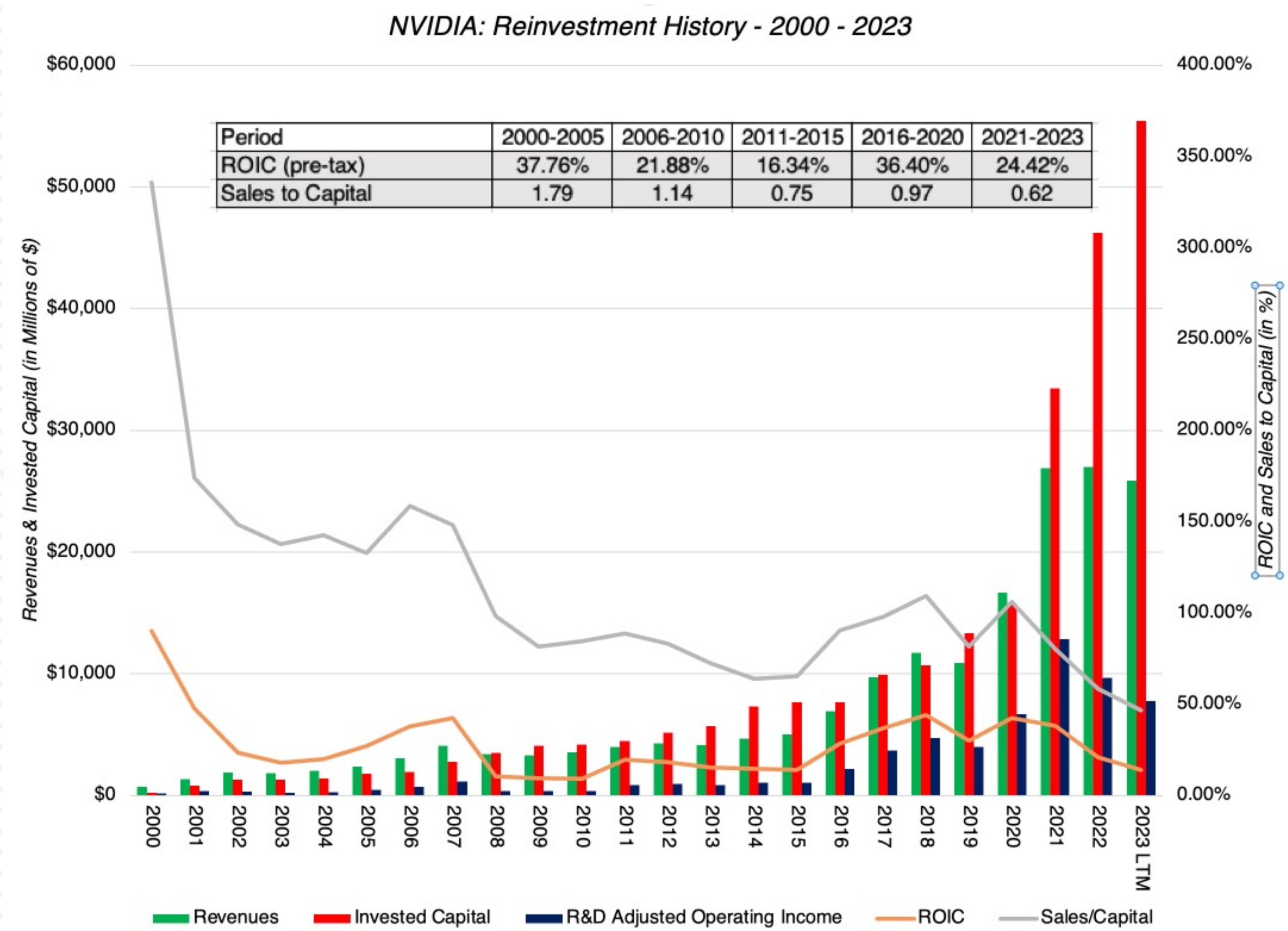




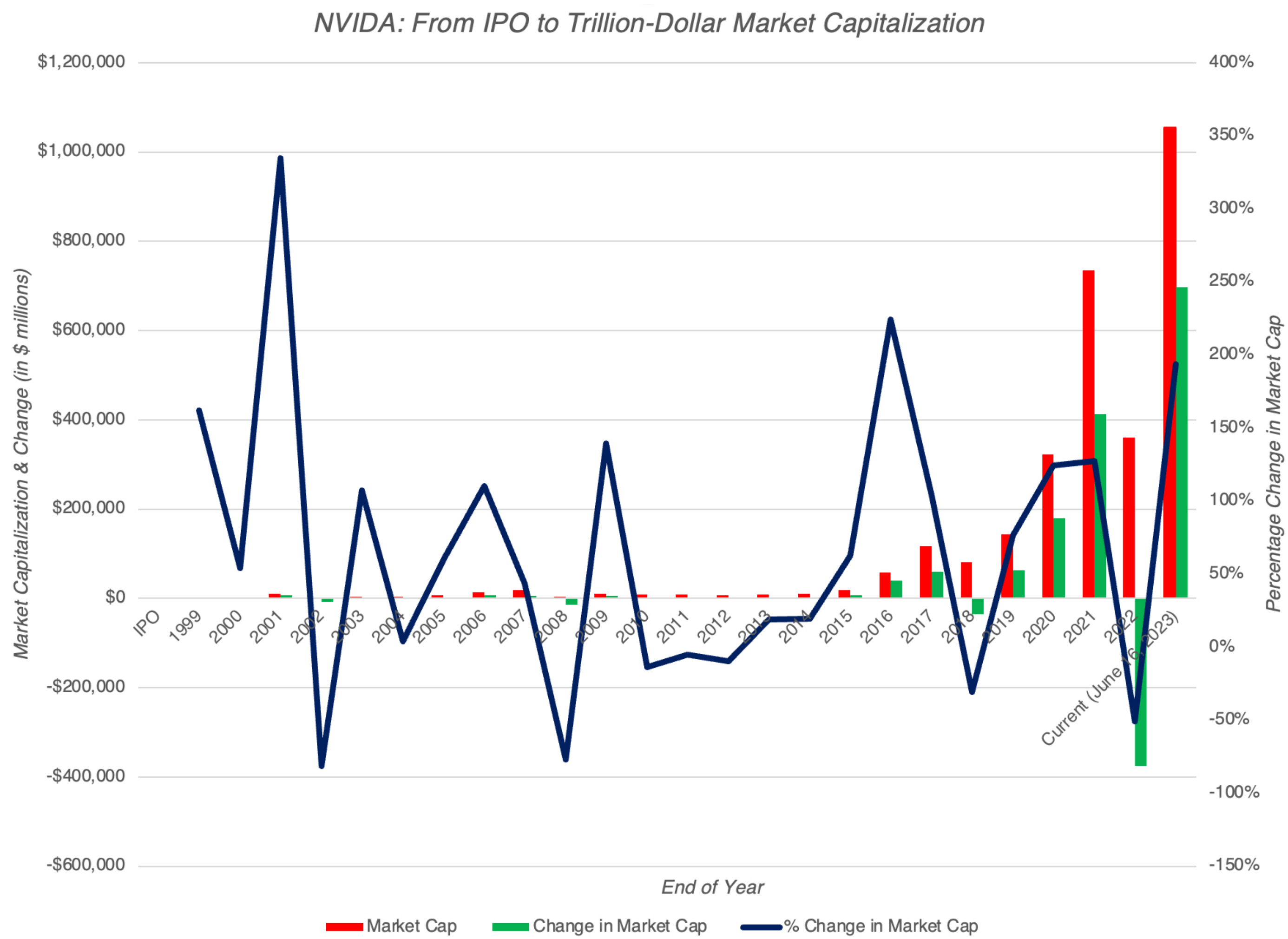
NVIDIA
1. Opportunistic Growth



2. With Large (but Productive) Reinvestment!



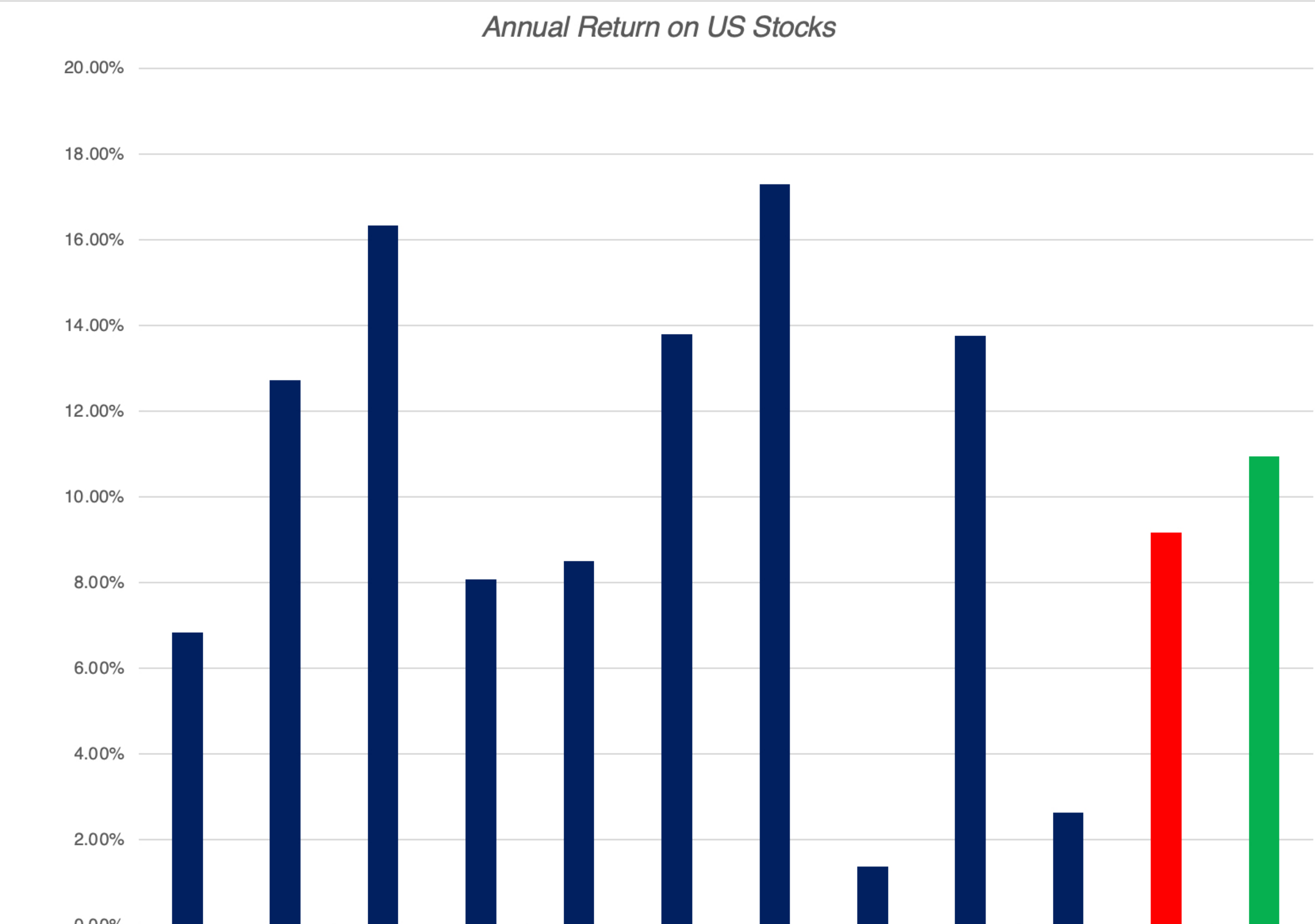
3. And a Mega Market Payoff



AI: Revolutionary or Incremental Change

- Revolutionary Changes not only affected wide swathes of businesses, some positively and some adversely, but that they also changed the ways that we live, work and interact.
- In parallel, we have also seen changes that are more incremental, and while significant in their capacity to create new businesses and disruption, don't quite qualify as revolutionary.
- So, where does AI fall on this spectrum from revolutionary to incremental to minimalist change? A year ago, I would have put it in the incremental column, but ChatGPT has changed my perspective. That was not because ChatGPT was at the cutting edge of AI technology, which it is not, but because it made AI relatable to everyone.

AI: Net Plus or Minus for Markets?



	1931-40	1941-50	1951-60	1961-70	1971-80	1981-90	1991-2000	2001-10	2011-20	2021-22	1931-1980	1981-2022
Annual Return	6.83%	12.72%	16.33%	8.08%	8.50%	13.80%	17.30%	1.38%	13.75%	2.63%	9.17%	10.95%

AI: Disruption's Dark Side

- The market is littered with the carcasses of what used to be successful businesses that have been disrupted by technological change.
- Investors in these disrupted companies not only lose money, as they get disrupted, but worse, invest even more in them, drawn by their "cheapness".
  - This happened, just to provide two examples, with investors in the brick-and-mortar retail companies that were devastated by online retail, and with investors in the newspaper/traditional ad companies that were upended by online advertising.
- If AI succeeds in its promise, will there be businesses that are upended and disrupted? Of course, but we are in the hype phase, where much more will be promised than can be delivered, but the biggest targets will come into focus sooner rather than later.

AI: Value Effects

- Hardware and Infrastructure: The AI effect on NVIDIA comes from the increased demand for AI-optimized computer chips, and as that market is expected to grow exponentially, the companies that can grab a large share of this market will benefit. There are undoubtedly other investments in infrastructure that will be needed to make the AI promise a reality.
- Software: AI hardware, by itself, has little value unless it is twinned with software that can take advantage of that computing power. This software can take multiple forms, from AI platforms, chatbots, deep learning algorithms (including image and voice recognition, as well as natural language processing) and machine learning.
- Data: Big data, used more as a buzzword than a business proposition, over the last decade may finally find its place in the value chain, when twinned with AI, but that pathway will not be linear or predictable.
- Applications: For companies that are more consumers of AI than its purveyors, the promise of AI is that it will change the way they do business, with positive and negative implications. The biggest pluses of AI, at least as presented by its promoters, is that it will allow companies to reduce costs (primarily by replacing manual labor with AI-driven applications) and make them more efficient, and by extension, more profitable. If every company has AI, and AI reduces costs and increases efficiency as promised for all of them, it is far more likely that they will end up with lower prices for their products/services and not higher profits.

AI: Social Effects

- I know that there are some advocates of AI who paint a picture of goodness, where AI takes over the menial tasks that presumably cause us boredom and brings an unbiased eye to data analysis that lead to better decisions. I know that there are others who see AI as an instrument that big companies will use to control minds and acquire power. With the experience of the big changes that have engulfed us in the last few decades still fresh, I would argue that they are both right.
- There are some who believe that AI can be held in check and made to serve its more noble impulses, by restricting or regulating its development, but I am not as optimistic for many reasons.
  - I believe that both regulators and legislators are woefully incapable of understanding the mechanics of AI, let alone pass sensible restrictions on its usage.
  - Second, any regulation or law that is aimed at preventing AI's excesses will almost certainly set in motion unintended consequences, that at least in some cases will be worse than the problems that the regulation/law was supposed to hold in check.
- While this is a pessimistic take, I believe that it a realistic one, and that just as with social media, it will be up to us, as consumers of AI products and services, to try to draw lines and separate good from bad. We may not succeed, but what choice do we have, but to try?

The AI Chip Story

- The AI story has particular resonance with NVIDIA because unlike most other companies, where it is mostly hand-waving about potential, it has substance in place already and a market that is its target. In particular, NVIDIA has spent much of the last few years investing and developing products for a nascent AI market. This lead time has given NVIDIA not just market leadership, but revenues and profits already. Much of the excited reaction to NVIDIA's most recent earnings report came from the company reporting a surge in its data center revenues, with much of the increase coming from AI chips.
- While the company does not explicitly break out how much of the data center revenues are from AI chips, it is estimated that the total market for those chips in 2022 was about $15 billion, with NVIDIA holding a dominant market share of about 80%. If those estimates are right, the bulk of the data center revenues for NVIDIA in 2022, which amounted to $15 billion in all, comes from AI-optimized chips.
- The ChatGPT jolt to market expectations has played out in increases in expected growth of the AI chip market over the next decade, with estimates for the overall AI chip market in 2030 ranging from $200 billion at the low end to close to $300 billion at the high end.

NVIDIA: The Lead In to the Story

- The driver of NVIDIA's success has been its high-performance GPU cards, but it is very likely that the businesses that bought these cards and drove NVIDIA's success in the last decade will be different from the businesses that will make it successful in the next one.
- For much of the last decade, it was gaming and crypto users that allowed the company to set itself apart from the competition, but the bad news is that both of these markets are maturing, with lower expected growth in the future.
- The good news, for NVIDIA, is that it has two other businesses that are ready to step in and contribute to growth.
  - The first is AI, where NVIDIA commands a hefty market share of what is now a relatively small market, but one that is almost certain to grow ten-fold or greater over the decade.
  - The other is in the automobiles business, where more powerful computing is seen as the ingredient needed to open up automated driving and other enhancements. NVIDIA is only a small player in this space, and while it does not enjoy the dominance that it does in AI, a growing market will allow NVIDIA to acquire a significant market share.

The NVIDIA Growth Story

- Revenue Growth: NVIDIA will remain a high growth company for two reasons.
  - The first is that in spite of its scaling up due to growth over the last decade, at least in terms of revenues, it has a modest market share of the overall semiconductor market, with revenues that are less than half of the revenues posted by Intel or TSMC.
  - The second, and more important reason, is that while its gaming revenue growth is starting to flag, it is well-positioned in AI and Auto, two markets poised for rapid growth. In my story, I will assume that these markets will deliver on their growth promise and that NVIDIA will maintain a dominant, albeit lower, market share of the AI chip business, while gaining a significant share (15%) of the Auto chip business

	AI		*Auto*		*Gaming & Other*		NVIDIA	
	Current	*In 2033*	*Current*	*In 2033*	*Current*	*In 2033*	*Current*	*In 2033*
Total Market ($ Mil)	$15,000	$325,000	$20,000	$200,000	Revenue growth: 15% for years 1-5, declining thereafter.		Revenues increase 10-fold over next decade	
Market Share	75%	60%	3.00%	15%				
NVIDIA revenues ($ Mil)	$11,250	$195,000	$600	$30,000	$14,028	$41,672	$25,878	$266,672

NVIDIA: The Rest of the Story

- Profitability: The semiconductor business has a cost structure that has relatively little flex to it, but I will assume in my NVIDIA story that the right margin to focus on is the R&D adjusted version, and that NVIDIA will bounce back quickly from its 2022 margin setback to deliver higher margins than its peer group. While my target R&D adjusted margin of 40% may look high, it is worth remembering that the company delivered 42.5% as margin in 2020 and 38.4% as margin in 2021.
- Investment Efficiency: NVIDIA has invested heavily in the last decade, generating only 65 cents in revenues for every dollar of capital invested (including the investment in R&D), in 2022. I believe that given the company's larger scale, with the payoff from past investments augmenting revenues, the company's sales to invested capital will approach the global industry median, which is $1.15 in revenues for every dollar of capital invested.
- Risk: I estimated NVIDIA's cost of capital based upon its geographic exposure and very low debt ratio to be 13.13%, but chose to use the industry average for US semiconductor companies, which was 12.21%, as the cost of capital in the initial growth period. Over time, I will assume that this cost of capital will drift down towards the overall market average cost of capital of 8.85%.

Nvidia	Jun-23

Base Year and Comparison

	Company	Industry
Revenue Growth	-15.29%	8.81%
Revenue	$14,028	
Operating Margin	25.64%	22.96%
Operating Income	$3,597	
EBIT (1-t)	$3,238	

Growth Story (Revenue)

Even as gaming and other chip businesses mature, NVIDIA"s investments in the AI and Auto chip businesses will deliver healthy growth over the next decade.

Profitability Story (Margin)

Margins improve, first as NVIDIA rebounds from a down-year and then long term, because of competitive advantages, especially in their growth businesses and economies of scale.

Growth Efficiency Story

Sales to capital moves to industry median, as company is able to slow its investment pace, and see payoffs from past investments in AI & Auto.

Terminal Value	
Growth Rate	3.60%
Cost of capital	8.85%
Return on capital	20.00%
Reinvestment Rate	18.00%

Value of Rest	$106,978
Value of AI	$417,459
Value of Auto	$58,568
Probability of failure =	0.00%
Value of operating assets =	$583,005
- Debt	$12,080
- Minority interests	$0
+ Cash	$15,320
+ Non-operating assets	$505
Value of equity	$586,750
- Value of options	$0
Value of equity in common sto	$586,750
Number of shares	2,470.00
Estimated value /share	**$237.55**
Price per share	$409.00
% Under or Over Valued	72.17%

	1	2	3	4	5	6	7	8	9	10	Terminal year
Revenue (Gaming/Other)	$ 16,132	$ 18,552	$ 21,335	$ 24,535	$ 28,215	$ 31,804	$ 35,125	$ 37,991	$ 40,225	$ 41,673	$ 43,173
Revenue (AI)	$ 29,253	$ 46,512	$ 63,027	$ 78,798	$ 93,825	$ 116,292	$ 137,643	$ 157,878	$ 176,997	$ 195,000	$ 202,020
Revenue (Auto)	$ 1,848	$ 3,672	$ 6,072	$ 9,048	$ 12,600	$ 15,504	$ 18,696	$ 22,176	$ 25,944	$ 30,000	$ 31,080
Revenues (Total)	$ 47,233	$ 68,736	$ 90,434	$ 112,381	$ 134,640	$ 163,600	$ 191,464	$ 218,045	$ 243,166	$ 266,673	$ 276,273
R&D Adj Operating Margin	35.00%	37.00%	38.00%	39.00%	40.00%	40.00%	40.00%	40.00%	40.00%	40.00%	40.00%
Operating Income	$ 16,532	$ 25,432	$ 34,365	$ 43,829	$ 53,856	$ 65,440	$ 76,585	$ 87,218	$ 97,266	$ 106,669	$ 17,269
EBIT (1-t)	$ 14,878	$ 22,889	$ 30,928	$ 39,446	$ 48,471	$ 56,933	$ 64,332	$ 70,647	$ 75,868	$ 80,002	$ 12,952
Reinvestment	$ 18,698	$ 18,868	$ 19,085	$ 19,356	$ 25,183	$ 24,229	$ 23,114	$ 21,844	$ 20,441	$ 8,348	$ 2,331
FCFF	$ (3,820)	$ 4,021	$ 11,844	$ 20,090	$ 23,288	$ 32,704	$ 41,218	$ 48,802	$ 55,427	$ 71,654	$ 10,621
										$ 58,568.24	
Cost of Capital	12.21%	12.21%	12.21%	12.21%	12.21%	11.54%	10.86%	10.19%	9.52%	8.85%	
Cumulated WACC	0.8912	0.7942	0.7078	0.6308	0.5622	0.5040	0.4546	0.4126	0.3767	0.3461	
Sales to Capital	1.15	1.15	1.15	1.15	1.15	1.15	1.15	1.15	1.15	1.15	
ROIC	37.16%	38.97%	39.85%	40.79%	41.77%	40.31%	38.88%	37.46%	36.06%	34.65%	20.00%

Risk Story

Initial cost of capital set equal to semiconductor industry average, for first 5 years, moving towards average for entire market over time.

Competitive Advantages

Strong competitive edges allow NVIDIA to earn well above its cost of capital for the next decade and beyond.

AI and Auto Business: Market Size and Market Share

	AI		*Auto*	
	Current	*In 2033*	*Current*	*In 2033*
Total Market ($ M)	$15,000	$325,000	$20,000	$200,000
Market Share	75%	60%	3.00%	15%
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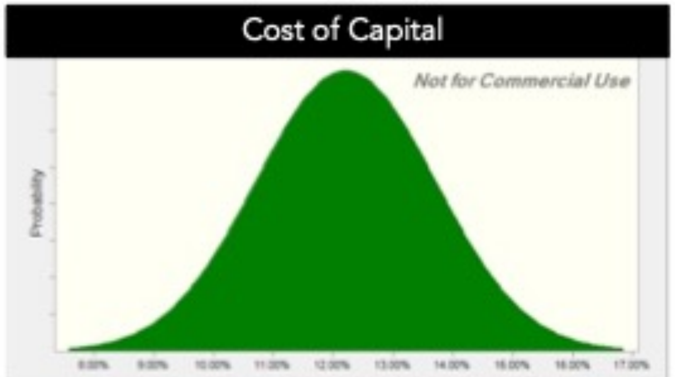
NVIDA: Value Simulation

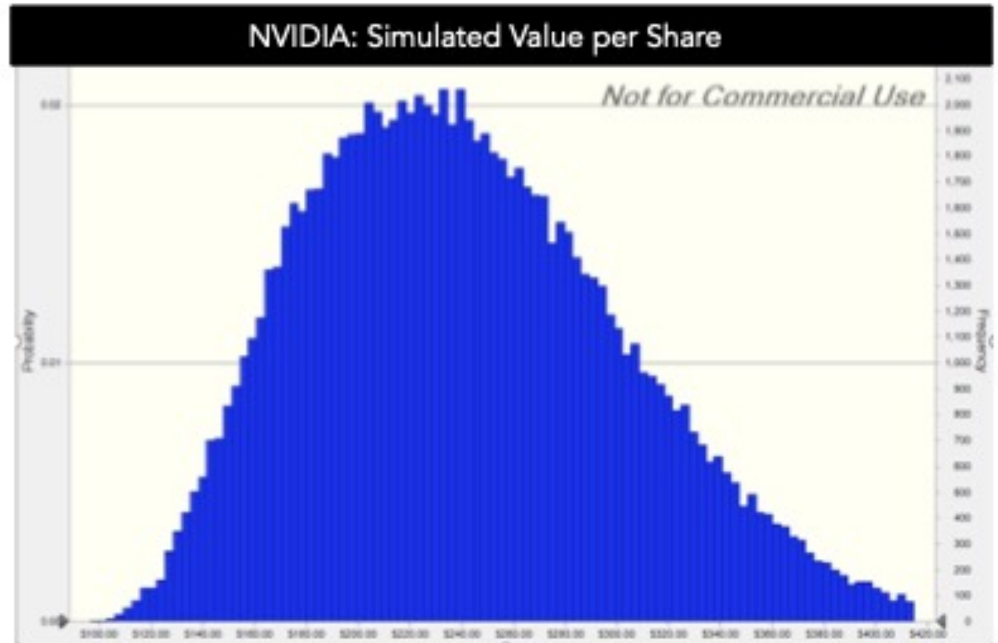
- Revenue Growth
  - Base Case: Revenues of $267 billion in 2033
  - Distribution:

	Base Case	Distribution	Rationale
AI: Total Market	$325 billiion	Uniform: $225 to $425 billion	Range of outside estimates
AI: NVIDIA Share	60%	Lognormal: Std Dev = 15%	More errors on upside than down
Auto: Total Market	$200 billion	Uniform: $100 - $300 billion	Range of outside estimates
Auto: NVIDIA Share	15%	Lognormal: Std Dev = 10%	More errors on upside than down
Gaming & Other: CAGR	15%	Triangular: 5% to 25%	Supplemental (to Gaming) business absence or presence

- Operating Margin
  - Base Case: Pre-tax operating margin of 40% (target)
  - Distribution: Triangular, with 30% (low) and 50% (high)
- Reinvestment
  - Base Case: Sales to Invested Capital of 1.15
  - Distribution: Lognormal, with range from 0.8-1.94
- Cost of Capital
  - Base Case: Industry average of 12.21%
  - Distribution: Normal Distribution (with standard dev of 1.5%)

NVIDIA: VALUE SIMULATION (JUNE 12, 2023)





Percentile	Value Per Share
0%	$96.51
10%	$166.01
20%	$186.59
30%	$203.88
40%	$220.23
50%	$236.28
60%	$252.97
70%	$271.77
80%	$294.04
90%	$326.17
100%	$555.75

And a Breakeven Analysis

		Revenues in 2033 (in billions)				
		$100.00	$200.00	$300.00	$400.00	$500.00
Operating Margin (%)	30%	$ 89.09	$ 137.19	$ 189.32	$ 244.03	$ 296.07
	35%	$ 104.24	$ 163.15	$ 226.88	$ 293.88	$ 357.71
	40%	$ 119.39	$ 189.61	$ 264.43	$ 343.73	$ 419.35
	45%	$ 134.54	$ 216.08	$ 301.99	$ 393.58	$ 480.99
	50%	$ 149.70	$ 242.55	$ 339.54	$ 443.43	$ 542.63

The Take-Aways from What ifs?

- The breakeven table reinforces the findings in the simulation, insofar as it shows that there are plausible paths that lead to the current price being a fair value or under value, but these paths require a daunting combination of extraordinary revenue growth and super-normal margins.
- In my view, a target margin of 50% is pushing the limits of possibility, in the semiconductor business, and if NVIDIA finds a way to deliver value that justifies current pricing, it has to be through explosive revenue growth.
- Put simply, you need another market or two, with potential similar to the AI market, where NVIDIA can wield a dominant market share to justify its pricing.

Judgment Day: Investing

- I love NVIDIA as a company and have nothing but praise for Jensen Huang's leadership of the company. My valuation story for NVIDIA reflects all of the positive features in the company will continue into the next decade, but that upbeat narrative still yields a value well below the current price.
- I would be lying if I said that selling one of my biggest winners is easy, especially since there is a plausible pathway, albeit a low-probability one, that the company will be able to deliver solid returns, at current prices.
  - I chose a path that splits the difference, selling half of my holdings and cashing in on my profits, and holding on to the other half, more for the optionality (that the company will find other new markets to enter in the next decade).
  - The value purists can argue, with justification, that I am acting inconsistently, given my value philosophy, but I am pragmatist, not a purist, and this works for me.
- It does open up an interesting question of whether you should continue to hold a stock in your portfolio that you would not buy at today's stock prices, and it is one that I will return to in a future post.

Judgment Day: Trading

	Investing	*Trading*
Determinants of level	Cash flows, growth and risk	Demand & Supply
Causes for change	Shifts in expected cash flows, growth and risk.	Changes in mood & momentum
Players	**Investors**, estimating value, and betting on price convergence.	**Traders**, playing momentum, or shifts in momentum
Key skills	1. Assess risk and value businesses 2. Find pricing catalysts.	1. Gauge market mood/momentum 2. Detect shifts in that mood.
Personal characteristics	(a) Long time horizon, (b) Patience and (c) immunity from peer pressure	(a) Flexible time horizon, (b) Gambling instincts and (c) Crowd Reading
Tools	Intrinsic Valuation (DCF), Forensic Accounting	Peer group pricing, Charts and technical indicators

Trading Follow Through

- If you are a trader, even if you believe that NVIDIA's value is well below its price, you may buy NVIDIA on the expectation that the stock will continue to rise, borne upwards by momentum or incremental information.
  - Given the strength of momentum as a market-driver, you may very well generate high returns over the next weeks, months or even years, and you should not let "value scolds" get in the way of your enjoyment of your winnings.
  - My only pushback would be against those who argue that momentum can carry a stock forward forever, since it is the gift that both gives and takes away.
- The strength of momentum in the rise in NVIDIA's stock price will be played out in the the opposite direction, when (not if) momentum shifts, and if you are trading NVIDIA, you should be working on indicators that give you early warning of those shifts, not worrying about value.

AI: Bottom Line

- Even if you buy into the argument that AI will change the ways that we work and play, it does not necessarily follow that investing in AI-related companies will yield returns. You can get the macro story right, but you need to also consider how that story plays out across companies to be able to generate returns.
- Refusing to make estimates or judgments about how AI will affect the fundamentals (cash flows, growth and risk) in a business, just because you face significant uncertainty, will not make that uncertainty go away, and instead will create a vacuum that will be filled by arbitrary AI premiums.
- As a society, it is unclear whether adding AI to the mix will make us better or worse off, since every big technological change seems to bring with it unintended consequences.
- All in all, I am considering asking ChatGPT to write this post for me, using my own language and history, and I am open to the possibility that it could do a better job than I can.